1. Item 4 (Most Relevant)

Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

📌 **Why is this the most important aspect?**
This problem contains multiple pieces of information that must be interpreted accurately:

- "9 kilometres long" refers to a **one-way distance**
- "18 km walk" refers to the **round trip**
- "Down at twice that speed" requires interpreting a speed relationship
- "Latest time he can begin" involves **backward time reasoning**

📌 **If the text is misread:**

- Treating 9 km as the round trip
- Ignoring the different speeds uphill and downhill
- Forgetting that 8 pm is the return time

➡️ The resulting mathematical model would be incorrect.

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2. Item 6

Identify and describe the mathematical aspects of a real-world problem situation, including identifying the significant variables

📌 **How this appears in the problem:**

- The context is a realistic trip-planning situation
- The mathematical components include:
  - Distance
  - Speed
  - Time
  - A fixed deadline (constraint)

📌 **Key competency:**

- Recognising this as a **distance–speed–time (DST)** problem
- Extracting the mathematical structure from the real-world context

3. Item 2

Identify the key variables in a model

📌 Key variables include:

- Uphill distance: 9 km
- Downhill distance: 9 km
- Uphill speed: 1.5 km/h
- Downhill speed: 3 km/h
- Time uphill and downhill
- Total walking time
- Starting time and return time (8 pm)

📌 There are more interacting variables than in the previous problem

4. Item 7

Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

📌 **Decomposition is crucial here:**

- Splitting the 18 km round trip into:
  - 9 km uphill
  - 9 km downhill
- Calculating each time separately and then adding them

📌 **Without this decomposition:**

- Students may incorrectly apply an average speed to the total distance

➡️ **This is a key cognitive challenge in the problem**

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5. Item 12

Identify the constraints, assumptions, and simplifications in a mathematical model

📌 **Explicit assumptions in the problem:**

- Speeds already include rest and meal breaks
- Speeds are constant uphill and downhill
- 8 pm is a strict deadline

📌 **Students must recognise that:**

- No additional time needs to be added for breaks
- The situation has already been simplified into an idealised model

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6. Item 9

Translate a problem into a standard mathematical representation or algorithm

📌 **The corresponding mathematical procedure:**

- Uphill time: $9 \div 1.5 = 6$ hours
- Downhill time: $9 \div 3 = 3$ hours
- Total time: $6 + 3 = 9$ hours

📌 **This forms a clear and logical calculation sequence**

7. Item 8

Recognise aspects of a problem that correspond with known mathematical concepts, facts, or procedures

📌 **Relevant mathematical concepts:**

- Speed = distance ÷ time
- Round-trip motion
- Piecewise motion
- Forward and backward reasoning on a time line

📌 **This is a familiar but cognitively demanding type of application problem**

8. Item 5

Recognise mathematical structure (including regularities, relationships, and patterns)

📌 **Structural features:**

- Same distance, different speeds → different times
- Additivity of time
- Using backward reasoning to determine the start time

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9. Item 1

Select a mathematical description or representation that describes a problem

📌 **Chosen representation:**

- A piecewise arithmetic model
- Deliberately avoiding an average-speed model (which would be incorrect)

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10. Item 3

Select a representation appropriate to the problem context

📌 **Lower relevance because:**

- The context naturally leads to algebraic and arithmetic representations
- There is little need to choose among multiple representations

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11. Items 10 and 11 (Least Relevant)

Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

📌 **Reason for low relevance:**

- No diagrams or advanced symbolic modelling are required
- The problem can be solved using basic calculations alone

① 第 4 条（最核心）

Read, decode and make sense of statements … to create a model of the situation

📌 **为什么仍然是最重要?**
这道题包含多个关键信息，必须被**准确理解**：

- “9 kilometres long” 是 **单程**
- “18 km walk” 指的是 **往返**
- “down at twice that speed” 需要进行速度转换
- “latest time he can begin“ 是一个 **逆向时间推理问题**

📌 **如果读错：**

- 把 9 km 当成往返
- 忽略上下山速度不同
- 忽略“8 pm 是返回时间”

➡️ 模型将完全错误

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② 第 6 条

Identify and describe the mathematical aspects of a real-world problem situation

📌 **体现：**

- 这是一个真实的登山行程规划问题
- 数学要素包括：
  - 距离
  - 速度
  - 时间
  - 截止时间（约束）

📌 **关键能力：**

- 看出这是一个“**路程–速度–时间（DST）**”问题
- 从生活情境中识别数学结构

③ 第 2 条

Identify the key variables in a model

📌 **关键变量包括：**

- 上山距离：9 km
- 下山距离：9 km
- 上山速度：1.5 km/h
- 下山速度：3 km/h
- 上山时间、下山时间
- 总用时
- 出发时间、返回时间（8 pm）

📌 **变量数量明显多于前一题**

④ 第 7 条

Simplify or decompose a situation or problem

📌 **分解思路非常关键：**

- 把“18 km 往返”拆成：
  - 上山 9 km
  - 下山 9 km
- 分别计算时间，再相加

📌 **如果不分解：**

- 学生容易错误地用平均速度 × 总路程

➡️ **这是本题的一个重要认知难点**

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⑤ 第 12 条

Identify the constraints, assumptions, simplifications in a mathematical model

📌 **明确给出的假设：**

- 速度已包含休息和吃饭时间（无需额外考虑）
- 上下山速度恒定
- 8 pm 是绝对截止时间

📌 **学生需要意识到：**

- 不需要再加“休息时间”
- 这是一个理想化模型

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⑥ 第 9 条

Translate a problem into a standard mathematical representation or algorithm

📌 **对应的数学表达：**

- 上山时间：$9 \div 1.5 = 6$ 小时
- 下山时间：$9 \div 3 = 3$ 小时
- 总时间：$6 + 3 = 9$ 小时

📌 **这是一个清晰的计算流程**

⑦ 第 8 条

Recognise aspects of a problem that correspond with known mathematical concepts

📌 **对应的数学知识：**

- 速度 = 路程 ÷ 时间
- 往返运动
- 分段运动问题
- 时间轴上的“向前 / 向后推算”

📌 **属于常见但综合性较强的应用题类型**

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⑧ 第 5 条

Recognise mathematical structure (relationships, patterns)

📌 **结构特征：**

- 同一路程，不同速度 → 时间不同
- 时间具有可加性
- 最终通过“倒推”确定起点时间

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⑨ 第 1 条

Select a mathematical description or representation

📌 **选择了：**

- 分段路程的算术模型
- 而不是平均速度模型（这一点本身具有区分度）

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⑩ 第 3 条

Select a representation appropriate to the problem context

📌 **相关性较低原因：**

- 题目自然指向用公式和算术
- 不涉及图像、表格等多种表示的权衡

⑪ 第 10 / 11 条（相关性最低）

Use mathematical tools (variables, symbols, diagrams)

📌 **原因：**

- 不需要画图或复杂符号
- 基础计算即可完成